

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
39EJ DEMINERALIZATION SYSTEM
DESIGN OF TANK FDN MKD 39EJ-TNK-001

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 2 of (15)
900707_GAU01_AG36_CED01_39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS	3
1. INTENT	4
2. EXECUTIVE SUMMARY	4
3. SCOPE	4
4. DESIGN REQUIREMENTS	4
4.1 Design Loads	4
5. CONCRETE FOUNDATIONS	5
5.1 Design	5
5.2 Materials	5
5.3 Foundations for Aboveground Tanks	5
5.4 Minimum Concrete Cover	5
5.5 Allowable Soil Bearing Capacity	5
6. DESIGN CALCULATION FOR FOUNDATION	6
References	11
ATTACHMENT 1 INPUT DRAWINGS	12

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 3 of (15)
900707_GAU01_AG36_CED01_39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
FDN	:	Foundation
MKD	:	Marked

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 4 of (15)
900707_GAU01_AG36_CED01_39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

1. INTENT

This document is to Prepare for design of **DEMINERALIZED WATER TANK (39EJ-TNK-001)** Tank Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarrah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in excel. For member strength calculation force values are directly taken from load data provided.

4. DESIGN REQUIREMENTS

4.1 Design Loads

All Design Loads are indicated in Doc. No. MD-216-39EJ-EG-ME-GAD- 2001 attached in Attachment 1. Loads are also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 5 of (15)
900707_GAU01_AG36_CED01_39303_OU100692610_R3	CALCULATION DEMINEALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Raft Foundation with a Concrete Ringwall.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall & Raft	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 375 kN/m² at 1.2m depth below EGL. (refer Table 5 - BH-14 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008)).

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 6 of (15)
900707_GAU01_AG36_CED01_ 39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

6. DESIGN CALCULATION FOR FOUNDATION

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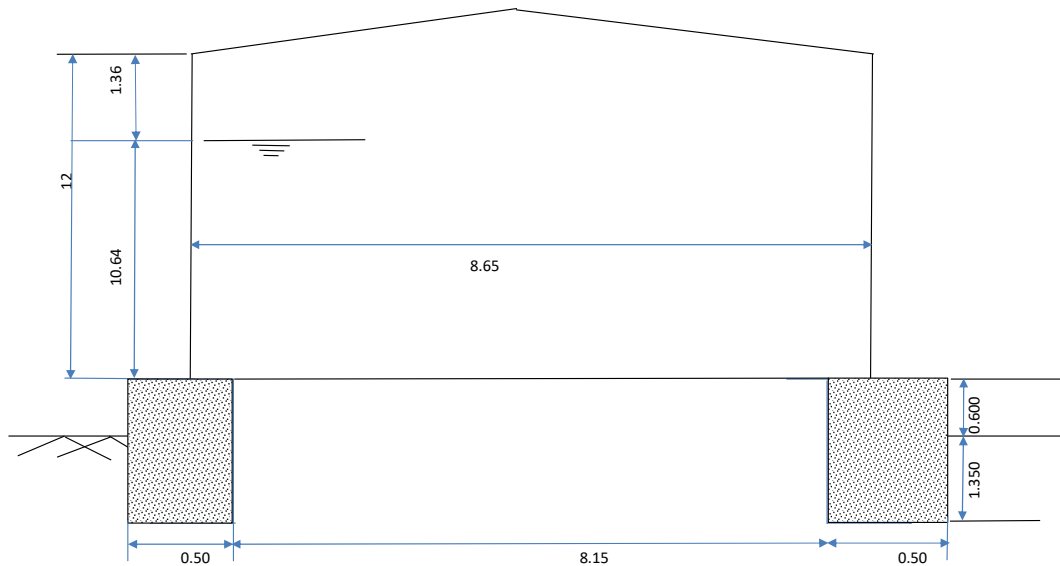
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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 7 of (15)
900707_GAU01_AG36_CED01_ 39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

Tank Loading Data

Weight, V	Loading Type		
	Empty (W)	383	kN
	Operating (Wo)	6808	kN
	Test (Wt)	7407	kN

Summary of Tank Specification



Diameter of Tank, D	=	8.65 m	
Height of Tank, H	=	12 m	
Height of Tank Fluid, H _L	=	10.64 m	
Inner Diameter of Concrete Ring Wall, ID	=	8.15 m	
Outer Diameter of Concrete Ring Wall, OD	=	9.15 m	
Height of Concrete Ring Wall, h	=	1.950 m	
ThickNess of Concrete Ring Wall, t	=	0.5 m	
B.C.D	=	13.75 m	
Top of Asphalt Board	=	0.6 m	(Above Finished Level)
Unit Weight of Liquid, γ_L	=	10 kN/m ³	
Unit Weight of Soil, γ_s	=	18 kN/m ³	
Unit Weight of Concrete, γ_c	=	24 kN/m ³	

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 8 of (15)
900707_GAU01_AG36_CED01_ 39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	59.00	624.00
	Operating	59.00	624.00
Seismic Loads	Operating	94.00	438.00

Foundation Weight

a) Ring Wall Weight = 635.89 kN

a) Inside Soil Weight = 1831.10 kN

Sectional Area

Net cross-sectional area for empty condition, A_e = 13.59 m²

Full cross-sectional area for operating condition, A_o = 65.76 m²

Sectional Modulus

Sectional modulus for empty condition, Z_e = 27.87 m³

Sectional modulus for operating condition, Z_o = 75.21 m³

Check of Minimum Wall ThickNess (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank, minimum thickNess of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + 0.6 \gamma_L H_L + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + 0.6 \gamma_L H_L + \gamma_c h$

or, $t = w / (0.4 \gamma_L * H_L + h(\gamma_s - \gamma_c))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D) = 14.09$ kN/m

So, Required Min Wall THK. = 0.457 m = 457 mm < 500 mm OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	6808.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	229.80	59.00	624.00
	Operating	LC103	D + F1 + W	6808.00	59.00	624.00
		LC104	D + F1 + 0.7E	6808.00	65.80	306.60
		LC105	0.6 (D + F1) + 0.7E	4084.80	65.80	306.60
	Test	LC106	D + F2	7407.00	0.00	0.00

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 9 of (15)
900707_GAU01_AG36_CED01_39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 375 kN/m²

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 498.75 kN/m²

Load Case	Loads at bottom of foundation level		$Q_{max} = V/A + M/Z$ kN/m ²	$Q_{max} = V/A - M/Z$ kN/m ²	Q_{all}/Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	9274.99	0.00	141.05	141.05	2.66	OK
LC102	865.69	739.05	90.23	37.19	5.53	OK
LC103	9274.99	739.05	150.88	131.23	3.31	OK
LC104	9274.99	434.91	146.84	135.27	3.40	OK
LC105	6551.79	434.91	105.42	93.86	4.73	OK
LC106	9873.99	0.00	150.16	150.16	2.50	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

$F_s = H$ resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	Hresist (kN)	Hacting (kN)	F_s	Sliding Check
	V (kN)					
LC101	9274.99	0.4	3710.00	0.00	-	-
LC102	865.69	0.4	346.28	59.00	5.9	OK
LC103	9274.99	0.4	3710.00	59.00	62.9	OK
LC104	9274.99	0.4	3710.00	65.80	56.4	OK
LC105	6551.79	0.4	2620.72	65.80	39.8	OK
LC106	9873.99	0.4	3949.60	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

$F_s = M$ resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	Mresist (kNm)	Macting (kNm)	F_s	Overturning Check
	V (kN)					
LC101	9274.99	4.575	42433.08	0.00	-	-
LC102	865.69	4.575	3960.53	739.05	5.4	OK
LC103	9274.99	4.575	42433.08	739.05	57.4	OK
LC104	9274.99	4.575	42433.08	434.91	97.6	OK
LC105	6551.79	4.575	29974.44	434.91	68.9	OK
LC106	9873.99	4.575	45173.51	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 , Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m ²)	Z (m ³)
Long term	Normal Operating	LC101	1.4 (D + F1)	9531.20	0.00	58.77	63.54
Short term	Empty	LC102	0.9 D + 1.6W	344.70	998.40	58.77	63.54
	Operating	LC103	1.2 (D + F1) + 1.6 W	8169.60	998.40	58.77	63.54
		LC104	1.2 (D + F1) + E	8169.60	438.00	58.77	63.54
		LC105	0.9 (D + F1) + E	6127.20	438.00	58.77	63.54
	Test	LC106	1.4 (D + F2)	10369.80	0.00	58.77	63.54

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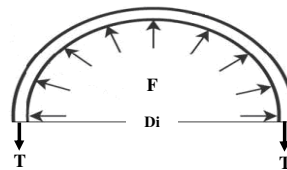
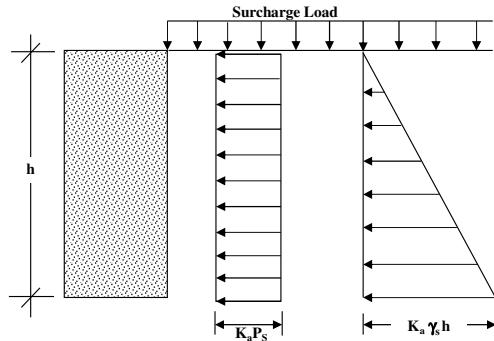
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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 10 of (15)
900707_GAU01_AG36_CED01_39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z kN/m ²
	V (kN)	M(kNm)	
LC101	9531.20	0.00	162.19
LC102	344.70	998.40	21.58
LC103	8169.60	998.40	154.73
LC104	8169.60	438.00	145.91
LC105	6127.20	438.00	111.16
LC106	10369.80	0.00	176.46

Maximum Surcharge Load, Ps
= 176.46 kN/m²



Where, K_a = coefficient of earth pressure at rest
= 0.5

$$P1 = K_a * P_s = 88.23 \text{ kN/m}^2$$

$$P2 = 1.4 * \gamma_s * h * K_a = 24.57 \text{ kN/m}^2 \quad \text{for LC106} \quad 1.4 (D + F2)$$

Maximum Lateral Load
 $F = P1 * h + 1/2 * P2 * h = 196.005 \text{ kN/m}$

Hoop Tension Load
 $T = 1/2 * F * ID = 798.721 \text{ kN}$

Hoop Tension Bar

1. Horizontal reinforcement

$$\phi = 0.9 \quad (\text{As per SBC 304, Table No. 21.2.2})$$

$$F_y = 420 \text{ MPa}$$

$$\text{Required } A_s = T / (\phi * F_y) = 2437.5 \text{ mm}^2$$

$$\text{Minimum } A_s = 0.0025 * b * D = 2437.5 \text{ mm}^2 \quad (\text{As per API 650, Cl. No. B.4.2.3})$$

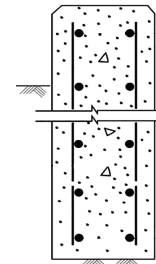
Hence, Provide 20 Nos. D16 = 3629.168 mm² OK

2. Vertical reinforcement

$$\text{Minimum } A_s = 0.0015 * b * D = 750 \text{ mm}^2 \quad (\text{As per API 650, Cl. No. B.4.2.3})$$

$$\text{So, Used } A_s = 2 \text{ row} * 6 \text{ ea} = 942 \text{ mm}^2 \quad (\text{D10 @ 167mm})$$

$$942 \text{ mm}^2 > \text{Required } A_s \quad \text{OK}$$



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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 11 of (15)
900707_GAU01_AG36_CED01_ 39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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MD-216-39EJ-EG-CV-CAL-1458	Mansourah Massarah Gold Project	Page : 12 of (15)
900707_GAU01_AG36_CED01_ 39303_OU100692610_R3	CALCULATION DEMINERALIZATION SYSTEM DESIGN OF TANK FDN MKD 39EJ-TNK-001	Date: 16.10.2023
O19045_450_C_K_24187_001_R3	AS BUILT	Rev: D01

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TANK ELEVATION
(SCALE 1:100)



SECTION C-C
SCALE- 1:20



PLATE ARRANGEMENT OF BOTTOM PLATE



SECTION F1-F1



DETAIL-1



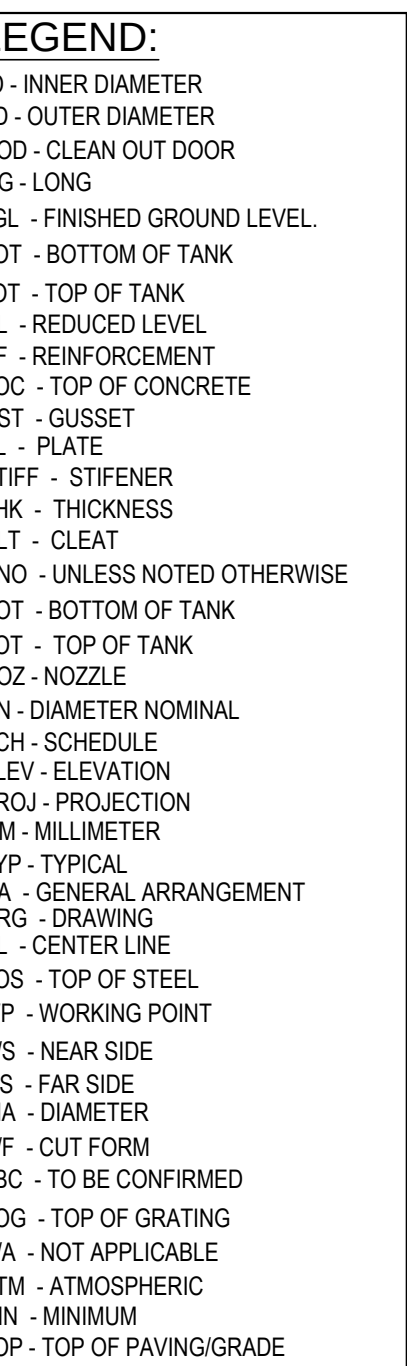
ROOF NOZZLES / FITTING



HELL NOZZLES / FITTINGS



LINING



LEGEND:

ID - INNER DIAMETER
OD - OUTER DIAMETER
CO - CLEAN OUT DOOR
CS - LONG
FG - FINISHED GROUND LEVEL
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
LR - REDUCED LEVEL
RE - REINFORCED
GCT - TOP OF CONCRETE
GST - GUSSET
PL - PLATE
STIFF - STIFFENER
TK - THICKNESS
CLT - CLEFT
UNO - UNLESS NOTED OTHERWISE
BOT - BOTTOM OF TANK
NOZ - NOZZLE
DN - DIAMETER NOMINAL
SCH - SCHEDULE
ELEV - ELEVATION
PROJ - PROJECTION
MM - MILLIMETER
TYP - TYPICAL
GA - GAGE ARRANGEMENT
DRG - DRAWING
CL - CENTER LINE
TOS - TOP OF STEEL
WP - WORKING POINT
NS - NEAR SIDE
FS - FACE SIDE
DA - DIAMETER
CF - CUT FORM
TBC - TO BE COMPLETED
TOG - TOP OF GRATING
N/A - NOT APPLICABLE
ATM - ATMOSPHERIC
MM - MINIMUM
TOP - TOP OF PAVING/GRADE



DETAIL OF CLEAN OUT DOOR (COD)



SECTION F-F



SECTION G1-G1

NOZZLE SCHEDULE													
NOZ	SERVICE	SIZE (DN)	SCH	RATING #	TYPE	ELEV "H"	PROJ "P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA *	RF PAD THK *	REMARK
N1	DEMIN WATER INLET	150	Sch XS	CL 150	RFSO	10.850	200	-	15°	SHELL	400	8	
N2	RECIRCULATION LINE FOR DEMIN WATER	150	Sch XS	CL 150	RFSO	10.850	200	-	C03 70°	SHELL	400	8	
N10	CO2 BREATHER	200	Sch XS	CL 150	RFSO	-	200	2205	37°	ROOF	485	8	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	200	50°	50°	SHELL	305	10	LOW TYPE RF PAD
N12	OVERFLOW	250	Sch XS	CL 150	RFSO	10.790	250	-	45°	SHELL	585	8	
N13	CLEAN OUT DOOR (COD) hdb	APR 1200	-	APR 650	-	-	-	-	180°	AP SHELL	2700/1372	12	
N14	SPARE C/W INLET	100	Sch XS	CL 150	RFSO	-	100	1800	127°	ROOF	305	6	
N20	DEMIN WATER OUTLET	200	Sch XS	CL 150	RFSO	0.400	200	-	80°	SHELL	485	10	
N21	DEMIN WATER OUTLET	200	Sch XS	CL 150	RFSO	0.400	200	-	100°	SHELL	485	10	
N22	DEMIN WATER OUTLET	200	Sch XS	CL 150	RFSO	0.400	200	-	120°	SHELL	485	10	
N23	DEMIN WATER OUTLET FOR REGENERATION PUMP	100	Sch XS	CL 150	RFSO	0.300	200	-	300°	SHELL	305	10	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	100	1800	157°	ROOF	305	6	
M1	MANHOLE C/W DAVIT	750	-	AP 1 650	-	0.925	175	-	270°	SHELL	1545	10	
M2	MANHOLE C/W HINGED COVER	600	-		-	-	500	3500	143°	ROOF	1150	6	

* REFER NOTE NO. 12

FOUNDATION LOAD DATA:					
BASIC LOAD CASE	VERTICAL COMPRESSION	VERTICAL UPLIFT	HORIZONTAL SHEAR	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT)	SLAB MOMENT
	(KN)	(KN)	(KN)	(KN-M)	(KN-M)
DEAD LOAD (NOMINAL THICKNESS)	383	—	—	—	—
DEAD LOAD (CORRODED THICKNESS)	320	—	—	—	—
ROOF PIPE LIQUID LOAD	53	—	—	—	—
SHELL PIPE LIQUID LOAD	53	—	—	—	—
LIVE LOAD	179	—	—	—	—
TANK LIQUID LOAD (OPERATING)	6140	—	—	—	—
TANK LIQUID LOAD (TEST)	6918	—	—	—	—
WIND LOAD ON SHELL	—	—	59	357	—
WIND LOAD ON ROOF	—	59	—	267	—
SEISMIC LOAD	213	213	94	438	556



KEY PLAN

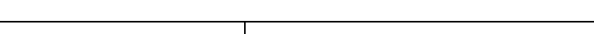
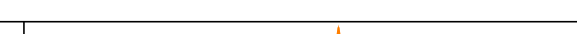
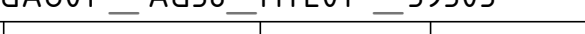
DESIGN DATA				
LOCATION		MANSOURAH - MASSARAH GOLD PROJECT		
TANK NAME		DEMATERIALIZED WATER TANK		
EQUIPMENT TAG No.		SPE-TM-001		
TANK DIMENSIONS		INSIDE DIAMETER: 8650mm, HEIGHT: 12000mm		
QUANTITY (NUMBER)		1		
ROOF TYPE & SLOPE		SLOPED CONE ROOF (SLOPE 1 IN 12) CONICAL APEX UP		
BOTTOM TYPE & SLOPE		SLOPED BOTTOM (SLOPE 1 IN 12) CONICAL APEX UP		
FIELD DICTATED/SHOP FABRICATED		FIELD DICTATED		
STORED FLUID		DEMATERIALIZED WATER		
DESIGN CODE	DESIGN CODE		API 650 TWELFTH EDITION, MARCH 2013. ADDENDUM 3, AUGUST 2018	
OPERATING DATA	OPERATING TEMPERATURE		48° C	
	NET WORKING CAPACITY		547 m ³	
	MAXIMUM CAPACITY		625 m ³	
	OPERATING LEVEL	MINIMUM	1010 mm	
MAXIMUM		10640 mm		
MECHANICAL & PROCESS DATA	OPERATING DENSITY OF CONTENT		1.01m ³	
	DESIGN TEMPERATURE		50° C	
	DESIGN PRESSURE (INTERNAL/EXTERNAL)		ATM/ATM	
	TEST PRESSURE		4.7M	
	HYDRO STATIC TEST PRESSURE		FULL OF WATER	
	DESIGN DENSITY OF CONTENT		1.01m ³	
	TESTING		AS PER API 650	
	INSPECTION		AS PER API 650	
	DENSITY OF SCALE		N/A	
		DESIGN SCALE THICKNESS		TANK SHELL/ROOF
DESIGN FILL RATE		150 m ³ /hr		
	DESIGN EMPTY RATE		380 m ³ /hr	
	CORROSION ALLOWANCE		SHELL	16 mm
			ROOF PLATE	0 mm
			BOTTOM PLATE / ANNUAL PLATE	16 mm
			ANCHOR BOLT	0 mm
STRUCTURAL	16 mm			
WIND LOAD DATA	BASIC WIND SPEED		43 m/sec. (155 Km/hr)	
	SITE EXPOSURE		C	
	VELOCITY PRESSURE EXPOSURE COEFFICIENT		K _e = AS PER ELEVATION	
	TOPOGRAPHIC FACTOR		K _z = 1.0	
	WIND DIRECTIONALITY FACTOR		K _d = 0.95	
	SITE CLASS		B	
	SEISMIC USER GROUP		SUG - I	
	IMPORTANCE FACTOR		(I) 1	
	SCALING FACTOR		(Q) 23	
	MAXIMUM DESIGN LIQUID LEVEL		(H) 10.659 m	
SEISMIC LOAD DATA	CO-EFFICIENT TO ADJUST THE SPECTRAL ACCELERATION FROM 5% TO 0.5% DAMPING		(K) 1.50	
	SEISMIC CO-EFFICIENTS		S=0.1g, S ₀ =0.02g S ₁ =1.0, T ₁ =1.0 S ₂ =0.007g, S ₃ =0.013g, T=4 sec.	
	ROOF LIFE LOAD		1.2kPa	
	FOUNDATION TYPE (RING WALL/SLAB)		RING WALL	
OTHER DATA	PWMT REQUIRED(YES/NO), IF YES SPECIFY REASON FOR PWMT		NO	
	REASON FOR PWMT(PROCESS REASON/PER API CODE)		AS PER API 650	
	RADIOGRAPHY		AS PER CL 8.1 API 650	
	(YES/NO)		NO	
	AGITATOR RISE REQUIRED		EQUIPMENT TYPE (AGITATOR/RAKE) NO AGITATOR/RAKE SUPPORTED BY TANK (YES/NO) N/A	
	INSULATION		SHELL	N/A
			ROOF	N/A
	LINER REQUIRED		(YES/NO)	YES
			LINER MATERIAL (NATURAL RUBBER (SHORE HARDNESS 65±5A)	
			LINER THICKNESS (ROOF/SHELL/BOLT)	0.6 mm
	PAINTING		YES	YES
			EXTERNAL	YES (UNLINED AREA)
	PIPE SUPPORT		YES	
	SPIRAL STAIRWAY		YES	
	ROOF CIRCUMFERENTIAL HANDRAIL		YES	
	WIND GRIDER		AS REQUIRED	
	CATHODIC PROTECTION		NO	
LEAK DETECTION (APPERTURANCE)		NO		
INTERNAL Baffles		NO		
EARTHING LUGS		YES		

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS AND FEET & DECIMALS & COORDINATES ARE IN METERS UNO.
2. ANCHOR BOLT MATERIAL SHALL BE A307 ASTM F 1554 GR. 36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. BOLT HEADS IN NOZZLE FLANGES & MANIFOLDS SHALL STRADDLE VERTICAL CENTER LINES ON NORTH & SOUTH SIDES CENTERING ON ROOF.
5. NOZZLE PROJECTION ON SHELL SHALL BE 50% RECESS FROM EXTERIOR SURFACE OF SHELL PLATE TO FACE OF FLANGE. NOZZLE PROJECTION ON ROOF ROOF REBERS FROM TOP OF ROOF PLATE TO FLANGE 300MM.
6. VERTICAL SHELL JOINT SHALL BE 100MM OFF SET BY 1000 MM.
7. CURVE ANGLE JOINTS SHALL ADJUST AT STAGER 100mm WITH VERTICAL SHELL PLATE WELD.
8. REINFORCEMENT PADS TO HAVE TELL TAIL THICKNESS OF 6 MM AND SHALL BE TESTED AT 100% PRESSURE WITH 100% WELD TENSILE AND 100% TENSILE TO WELD JOINTS.
9. FLANGE (CONE & MANIFOLD) GASKET SEALING SURFACE SHALL BE PROVIDED WITH 0.002 TO 0.003 (0.0002 TO 0.0012 IN) SURFACE FINISH.
10. EARTHING LOG LOCATION SHALL BE CONFIRMED IN SEPARATE DRAWING.
11. COLD GROOVE DIMENSIONS ARE FINALIZED BASED ON FOUNDATION RING WIDTH AND COLD BOTTOM PLATE DIMENSION.
12. STIFFENERS FOR NOZZLE IF REQUIRED SHALL BE INSTALLED ON TOP TO SHELL TO STIFFENERS.
13. WELDING SHALL CONFORM TO THE REQUIREMENTS OF THE WELDING CODE OF THE FABRICATOR WILL BE RELEASED FOR INSTALLATION OF LINER.
14. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING: 100-20-392-EGE-GAS-2002 & 2003.
15. WELDING SHALL CONFORM TO THE REQUIREMENTS OF THE WELDING CODE OF THE FABRICATOR (ASME SEC-9).
16. ALL NOZZLE FLANGES SHALL BE IN ACCORDANCE WITH ASME B16.5 FOR SUD UP TO DN600 AND TO ASME B16.4 FOR SIZES A FORD 6" & ABOVE.
17. SEE HANGAR REFER DRO. NO. MD-316-100-LOG-EST-DIED-1002-1003 (A1,2,3).



3D VIEW OF FOUNDATION NOTCH
(FOR COD)

														RESERVED FOR VENDOR LOGO AND DRAWING NUMBER														LARSEN & TOUBRO DRAWING NUMBER 019045_080_S_G_26089_001														REVISION 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